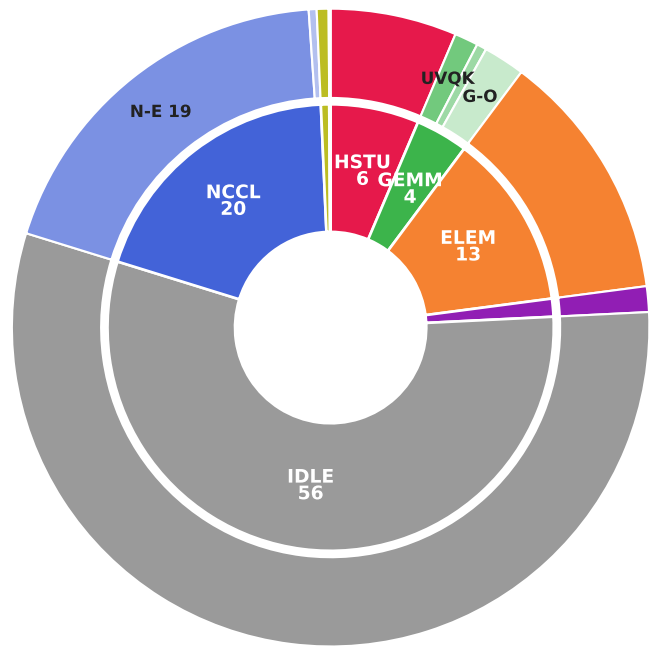
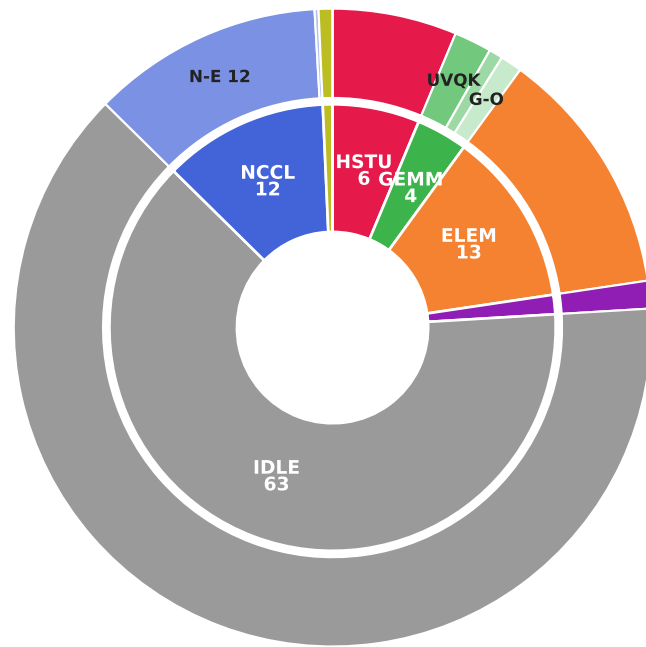


§8 scale-study GPU-time sunburst — GB300 16-GPU, exposed accounting, time-weighted aggregate over 80 steps
inner ring = grouping · outer ring = leaves · % of the step (exp4 = §5.2(B) fastest-step ref)

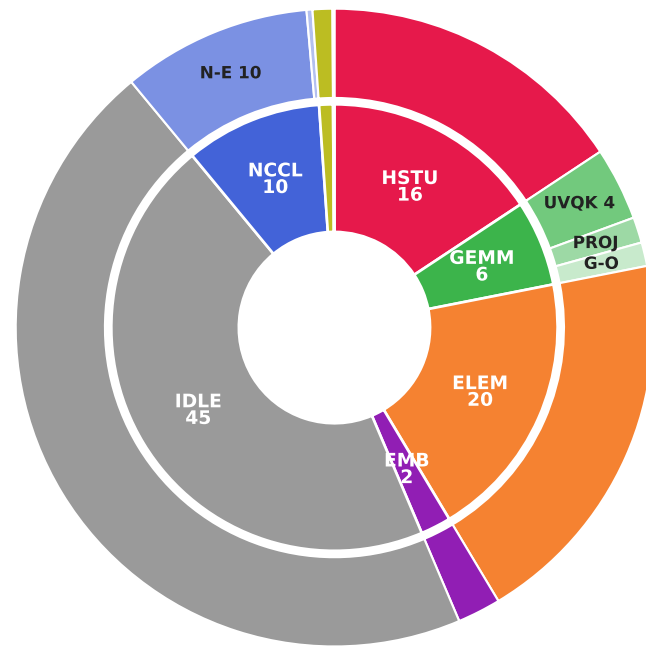
exp4 (ref, fastest) · 50M/2048



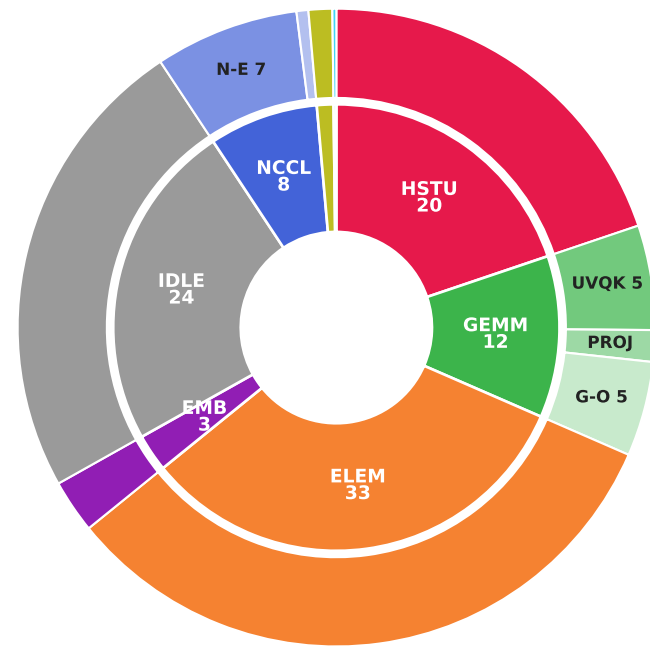
exp5 · 50M/2048 (80-step agg)



zipf · 1B/4096 (80-step agg)



lognormal · 1B/4096 (80-step agg)



■ HSTU = hstu fwd/bwd ■ UVQK = gemm / uvqk ■ PROJ = gemm / projection ■ G-O = gemm / others ■ ELEM = elementwise ■ EMB = embedding op ■ N-E = nccl exposed ■ N-O = nccl overlap ■ IDLE = GPU idle ■ OTH = others ■ OVL = overlapped